```
#ZAINA RAHEEN A P #241801328
#EX 2: DATA DISCOVERY AND EXPLORATORY DATA ANALYSIS
# TITANIC DATASET
# ============================================================
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.impute import SimpleImputer
from sklearn.model_selection import train_test_split
```

```
#ZAINA RAHEEN A P #241801328
# 1. Use titanic.csv and convert it into dataframe
# ------------------------------------------------------------

df = pd.read_csv("titanic.csv")
print("Titanic Dataset:")
print(df.head())
```

```
Titanic Dataset:
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

```
#ZAINA RAHEEN A P #241801328
# 2. Understand the data (shape, info, describe)
# ------------------------------------------------------------

print("\nShape of Dataset:")
print(df.shape)
print("\nDataset Information:")
print(df.info())
print("\nStatistical Description:")
print(df.describe())
```

```
Shape of Dataset:
(891, 12)

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None

Statistical Description:
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
```

```
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

```
#ZAINA RAHEEN A P #241801328
# 3. Fill missing values in Age using SimpleImputer
# ------------------------------------------------------------

imputer = SimpleImputer(strategy="mean")
df["Age"] = imputer.fit_transform(df[["Age"]]).ravel()
print("\nMissing Age values after imputation:")
print(df["Age"].isnull().sum())
```

```
Missing Age values after imputation:
0
```

```
#ZAINA RAHEEN A P #241801328
# 4. Fill missing Cabin with "Unknown"
# ------------------------------------------------------------

df["Cabin"] = df["Cabin"].fillna("Unknown")

print("\nMissing Cabin values:")
print(df["Cabin"].isnull().sum())
```

```
Missing Cabin values:
0
```

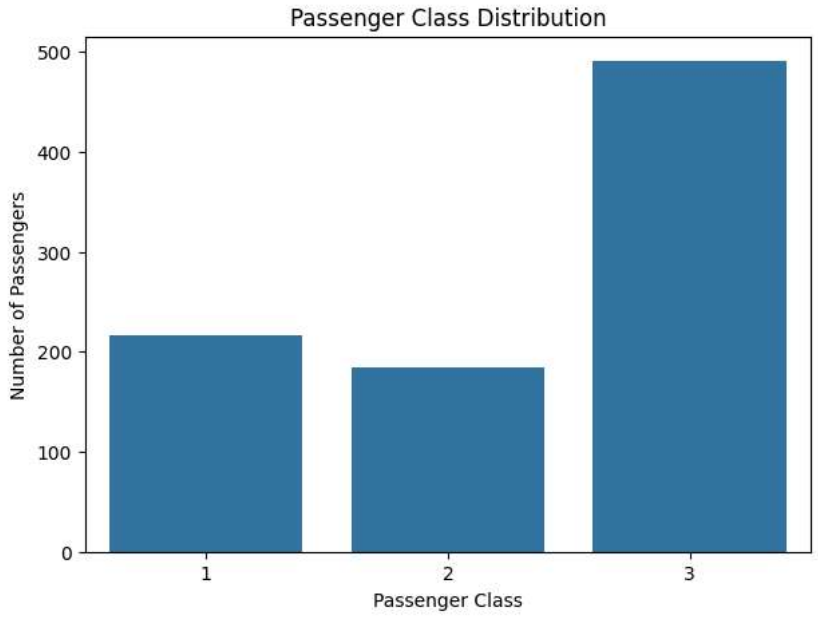
```
#ZAINA RAHEEN A P #241801328
# 5. Fill missing Embarked with the mode
# ------------------------------------------------------------

df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
print("\nMissing Embarked values:")
print(df["Embarked"].isnull().sum())
```

```
Missing Embarked values:
0
```

```
#ZAINA RAHEEN A P #241801328
# 6. Analyse Passenger Class using countplot
# ------------------------------------------------------------

plt.figure(figsize=(7, 5))
sns.countplot(x="Pclass", data=df)
plt.title("Passenger Class Distribution")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.show()
```




```
#ZAINA RAHEEN A P #241801328
# 7. List names of passengers who are female and survived
# ------------------------------------------------------------

female_survived = df[
    (df["Sex"] == "female") &
    (df["Survived"] == 1)
]
print("\n7. Female Passengers Who Survived:")
print(female_survived["Name"].to_string(index=False))
```

```
Swift, Mrs. Frederick Joel (Margaret Welles Bar...
                        Bystrom, Mrs. (Karolina)
                    Duran y More, Miss. Asuncion
  Beckwith, Mrs. Richard Leonard (Sallie Monypeny)
              Abelson, Mrs. Samuel (Hannah Wizosky)
                   Najib, Miss. Adele Kiamie "Jane"
     Potter, Mrs. Thomas Jr (Lily Alexenia Wilson)
      Shelley, Mrs. William (Imanita Parrish Hall)
                    Graham, Miss. Margaret Edith
```

```
#ZAINA RAHEEN A P #241801328
# 8. Find all passengers who are in 3rd class
#    and under the age of 18
# ------------------------------------------------------------

third_class_under18 = df[
    (df["Pclass"] == 3) &
    (df["Age"] < 18)
]

print("\n8. 3rd Class Passengers Under Age 18:")
print(third_class_under18[["Name", "Age", "Pclass"]].to_string(index=False))
```

```
8. 3rd Class Passengers Under Age 18:
                                          Name   Age  Pclass
                 Palsson, Master. Gosta Leonard  2.00       3
                Sandstrom, Miss. Marguerite Rut  4.00       3
           Vestrom, Miss. Hulda Amanda Adolfina 14.00       3
                           Rice, Master. Eugene  2.00       3
                   McGowan, Miss. Anna "Annie" 15.00       3
                  Palsson, Miss. Torborg Danira  8.00       3
                    Nicola-Yarred, Miss. Jamila 14.00       3
                     Panula, Master. Juha Niilo  7.00       3
             Goodwin, Master. William Frederick 11.00       3
                         Skoog, Master. Harald  4.00       3
                Andersson, Miss. Erna Alexandra 17.00       3
                     Goodwin, Miss. Lillian Amy 16.00       3
                         Ford, Mr. William Neal 16.00       3
                           Zabour, Miss. Hileni 14.50       3
                          Attalah, Miss. Malake 17.00       3
               Andersson, Miss. Ellis Anna Maria  2.00       3
                    Nicola-Yarred, Master. Elias 12.00       3
                              Osen, Mr. Olaf Elon 16.00       3
                Ford, Miss. Robina Maggie "Ruby"  9.00       3
                Gilnagh, Miss. Katherine "Katie" 16.00       3
                               Calic, Mr. Jovo 17.00       3
                    Panula, Master. Eino Viljami  1.00       3
Goldsmith, Master. Frank John William "Frankie"  9.00       3
                           Rice, Master. Arthur  4.00       3
                    Johnson, Miss. Eleanor Ileen  1.00       3
           Asplund, Master. Clarence Gustaf Hugo  9.00       3
             Kink-Heilmann, Miss. Luise Gretchen  4.00       3
                      Strom, Miss. Telma Matilda  2.00       3
                       Carr, Miss. Helen "Ellen" 16.00       3
                  Sunderland, Mr. Victor Francis 16.00       3
                  Asplund, Miss. Lillian Gertrud  5.00       3
               Asplund, Master. Edvin Rojj Felix  3.00       3
                        Panula, Mr. Ernesti Arvid 16.00       3
                               Rice, Master. Eric  7.00       3
                        de Pelsmaeker, Mr. Alfons 16.00       3
                 Vander Planke, Mr. Leo Edmondus 16.00       3
          Coutts, Master. William Loch "William"  3.00       3
                               Elias, Mr. Tannous 15.00       3
                     Palsson, Miss. Stina Viola  3.00       3
                    Nakid, Miss. Maria ("Mary")  1.00       3
                 Goodwin, Master. Sidney Leonard  1.00       3
                       Van Impe, Miss. Catharina 10.00       3
                      Kallio, Mr. Nikolai Erland 17.00       3
                   Baclini, Miss. Marie Catherine  5.00       3
                    Baclini, Miss. Helene Barbara  0.75       3
                         Hirvonen, Miss. Hildur E  2.00       3
                 Goodwin, Master. Harold Victor  9.00       3
           Coutts, Master. Eden Leslie "Neville"  9.00       3
                             Calic, Mr. Petar 17.00       3
                          Elias, Mr. Joseph Jr 17.00       3
            Andersson, Miss. Ingeborg Constanzia  9.00       3
               Andersson, Miss. Sigrid Elisabeth 11.00       3
                    Rush, Mr. Alfred George John 16.00       3
                              Skoog, Miss. Mabel  9.00       3
                   Skoog, Miss. Margit Elizabeth  2.00       3
```

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#ZAINA RAHEEN A P #241801328
# 9. Find passengers who are in class 1
#    and older than 40
# ------------------------------------------------------------

class1_over40 = df[
```

```
    (df["Pclass"] == 1) &
    (df["Age"] > 40)
]

print("\n9. Class 1 Passengers Older Than 40:")
print(class1_over40[["Name", "Age", "Pclass"]].to_string(index=False))
```

```
9. Class 1 Passengers Older Than 40:
                                                Name  Age  Pclass
                                 McCarthy, Mr. Timothy J 54.0       1
                                Bonnell, Miss. Elizabeth 58.0       1
                             Holverson, Mr. Alexander Oskar 42.0       1
                   Harper, Mrs. Henry Sleeper (Myna Haxtun) 49.0       1
                             Ostby, Mr. Engelhart Cornelius 65.0       1
                                Harris, Mr. Henry Birkhardt 45.0       1
                                Chaffee, Mr. Herbert Fuller 46.0       1
                                  Goldschmidt, Mr. George B 71.0       1
                             Porter, Mr. Walter Chamberlain 47.0       1
                                White, Mr. Percival Wayland 54.0       1
                                Williams, Mr. Charles Duane 51.0       1
                                  Van der hoef, Mr. Wyckoff 61.0       1
                                    Smith, Mr. James Clinch 56.0       1
                                 Isham, Miss. Ann Elizabeth 50.0       1
          Romaine, Mr. Charles Hallace ("Mr C Rolmane") 45.0       1
              Brown, Mrs. James Joseph (Margaret Tobin) 44.0       1
                                      Lurette, Miss. Elise 58.0       1
                              Minahan, Dr. William Edward 44.0       1
                                Stead, Mr. William Thomas 62.0       1
                                         Taussig, Mr. Emil 52.0       1
             Graham, Mrs. William Thompson (Edith Junkins) 58.0       1
                          Andrews, Miss. Kornelia Theodosia 63.0       1
       Baxter, Mrs. James (Helene DeLaudeniere Chaput) 50.0       1
                                   Partner, Mr. Austen 45.5       1
                         Burns, Miss. Elizabeth Margaret 41.0       1
                            Blackwell, Mr. Stephen Weart 45.0       1
      Warren, Mrs. Frank Manley (Anna Sophia Atkinson) 60.0       1
                                  Bidois, Miss. Rosalie 42.0       1
                              Silvey, Mr. William Baird 50.0       1
                                       Fortune, Mr. Mark 64.0       1
                           Peuchen, Major. Arthur Godfrey 52.0       1
                                 Goldenberg, Mr. Samuel L 49.0       1
                              Millet, Mr. Francis Davis 65.0       1
                                    Anderson, Mr. Harry 48.0       1
                                      Gee, Mr. Arthur H 47.0       1
                                Smart, Mr. John Montgomery 56.0       1
                                   Kent, Mr. Edward Austin 58.0       1
                                Molson, Mr. Harry Markland 55.0       1
                                   Artagaveytia, Mr. Ramon 71.0       1
                           Eustis, Miss. Elizabeth Mussey 54.0       1
        Rothschild, Mrs. Martin (Elizabeth L. Barrett) 54.0       1
                             Walker, Mr. William Anderson 47.0       1
       Hippach, Mrs. Louis Albert (Ida Sophia Fischer) 44.0       1
                          Butt, Major. Archibald Willingham 45.0       1
                                 Douglas, Mr. Walter Donald 50.0       1
                              Nicholson, Mr. Arthur Ernest 64.0       1
                                       Wright, Mr. George 62.0       1
Duff Gordon, Lady. (Lucille Christiana Sutherland) ("Mrs Morgan") 48.0       1
            Appleton, Mrs. Edward Dale (Charlotte Lamson) 53.0       1
                       Frolicher-Stehli, Mr. Maxmillian 60.0       1
       Stephenson, Mrs. Walter Bertram (Martha Eustis) 52.0       1
          Duff Gordon, Sir. Cosmo Edmund ("Mr Morgan") 49.0       1
                            Kimball, Mr. Edwin Nelson Jr 42.0       1
                                   Sutton, Mr. Frederick 61.0       1
                     Barkworth, Mr. Algernon Henry Wilson 80.0       1
```

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#ZAINA RAHEEN A P #241801328
# 10. Find passengers survived in the above category
#     (Class 1 and Age > 40)
# ------------------------------------------------------------

class1_over40_survived = df[
    (df["Pclass"] == 1) &
    (df["Age"] > 40) &
    (df["Survived"] == 1)
]

print("\n10. Class 1 Passengers Older Than 40 Who Survived:")
print(class1_over40_survived[
    ["Name", "Age", "Pclass", "Survived"]
].to_string(index=False))
```

```
10. Class 1 Passengers Older Than 40 Who Survived:
                                                Name  Age  Pclass  Survived
                                Bonnell, Miss. Elizabeth 58.0       1         1
                   Harper, Mrs. Henry Sleeper (Myna Haxtun) 49.0       1         1
          Romaine, Mr. Charles Hallace ("Mr C Rolmane") 45.0       1         1
```

```
                          Brown, Mrs. James Joseph (Margaret Tobin) 44.0       1         1
                                                  Lurette, Miss. Elise 58.0       1         1
                       Graham, Mrs. William Thompson (Edith Junkins) 58.0       1         1
                                     Andrews, Miss. Kornelia Theodosia 63.0       1         1
                     Baxter, Mrs. James (Helene DeLaudeniere Chaput) 50.0       1         1
                                       Burns, Miss. Elizabeth Margaret 41.0       1         1
                  Warren, Mrs. Frank Manley (Anna Sophia Atkinson) 60.0       1         1
                                            Bidois, Miss. Rosalie 42.0       1         1
                                     Peuchen, Major. Arthur Godfrey 52.0       1         1
                                           Goldenberg, Mr. Samuel L 49.0       1         1
                                              Anderson, Mr. Harry 48.0       1         1
                                     Eustis, Miss. Elizabeth Mussey 54.0       1         1
                       Rothschild, Mrs. Martin (Elizabeth L. Barrett) 54.0       1         1
                     Hippach, Mrs. Louis Albert (Ida Sophia Fischer) 44.0       1         1
Duff Gordon, Lady. (Lucille Christiana Sutherland) ("Mrs Morgan") 48.0       1         1
                      Appleton, Mrs. Edward Dale (Charlotte Lamson) 53.0       1         1
                                   Frolicher-Stehli, Mr. Maxmillian 60.0       1         1
                  Stephenson, Mrs. Walter Bertram (Martha Eustis) 52.0       1         1
                     Duff Gordon, Sir. Cosmo Edmund ("Mr Morgan") 49.0       1         1
                                       Kimball, Mr. Edwin Nelson Jr 42.0       1         1
                               Barkworth, Mr. Algernon Henry Wilson 80.0       1         1
                                    Harper, Mr. Henry Sleeper 48.0       1         1
                                Simonius-Blumer, Col. Oberst Alfons 56.0       1         1
                                     Frauenthal, Dr. Henry William 50.0       1         1
                                  Calderhead, Mr. Edward Pennington 42.0       1         1
                                           Taylor, Mr. Elmer Zebley 48.0       1         1
                               Hogeboom, Mrs. John C (Anna Andrews) 51.0       1         1
            Robert, Mrs. Edward Scott (Elisabeth Walton McMillan) 43.0       1         1
                                    Leader, Dr. Alice (Farnham) 49.0       1         1
               Hays, Mrs. Charles Melville (Clara Jennings Gregg) 52.0       1         1
                         Stone, Mrs. George Nelson (Martha Evelyn) 62.0       1         1
                        Wick, Mrs. George Dennick (Mary Hitchcock) 45.0       1         1
                                         Daly, Mr. Peter Denis  51.0       1         1
             Swift, Mrs. Frederick Joel (Margaret Welles Barron) 48.0       1         1
                Beckwith, Mrs. Richard Leonard (Sallie Monypeny) 47.0       1         1
                     Potter, Mrs. Thomas Jr (Lily Alexenia Wilson) 56.0       1         1
```

```
#ZAINA RAHEEN A P #241801328
# 11. Which male passengers paid a fare greater than 100?
# ------------------------------------------------------------

male_fare100 = df[
    (df["Sex"] == "male") &
    (df["Fare"] > 100)
]
print("\n11. Male Passengers Who Paid Fare > 100:")
print(male_fare100[
    ["Name", "Fare", "Sex"]
].to_string(index=False))
```

```
11. Male Passengers Who Paid Fare > 100:
                                    Name     Fare  Sex
            Fortune, Mr. Charles Alexander 263.0000 male
                  Baxter, Mr. Quigg Edmond 247.5208 male
            Allison, Master. Hudson Trevor 151.5500 male
                 Graham, Mr. George Edward 153.4625 male
                       Ringhini, Mr. Sante 135.6333 male
                 Widener, Mr. Harry Elkins 211.5000 male
                Carter, Mr. William Ernest 120.0000 male
                          Fortune, Mr. Mark 263.0000 male
Penasco y Castellana, Mr. Victor de Satode 108.9000 male
                        Farthing, Mr. John 221.7792 male
               Douglas, Mr. Walter Donald 106.4250 male
              Thayer, Mr. John Borland Jr 110.8833 male
                      Robbins, Mr. Victor 227.5250 male
               Newell, Mr. Arthur Webster 113.2750 male
             Frauenthal, Dr. Henry William 133.6500 male
       Cardeza, Mr. Thomas Drake Martinez 512.3292 male
                  Thayer, Mr. John Borland 110.8833 male
                    Lesurer, Mr. Gustave J 512.3292 male
      Carter, Master. William Thornton II 120.0000 male
```

```
#ZAINA RAHEEN A P #241801328
# 12. List passengers who embarked from port C
#     and were in Class 2
# ------------------------------------------------------------

portC_class2 = df[
    (df["Embarked"] == "C") &
    (df["Pclass"] == 2)
]
print("\n12. Passengers Embarked from C and in Class 2:")
print(portC_class2[
    ["Name", "Embarked", "Pclass"]
].to_string(index=False))
```

```
12. Passengers Embarked from C and in Class 2:
                                                Name Embarked  Pclass
                  Nasser, Mrs. Nicholas (Adele Achem)        C       2
             Laroche, Miss. Simonne Marie Anne Andree        C       2
                               Nasser, Mr. Nicholas        C       2
                                 Richard, Mr. Emile        C       2
                                   Pernot, Mr. Rene        C       2
                               Levy, Mr. Rene Jacques        C       2
                                  Abelson, Mr. Samuel        C       2
                            del Carlo, Mr. Sebastiano        C       2
                                Lehmann, Miss. Bertha        C       2
         Jerwan, Mrs. Amin S (Marie Marthe Thuillard)        C       2
                           Padro y Manent, Mr. Julian        C       2
Laroche, Mrs. Joseph (Juliette Marie Louise Lafargue)        C       2
                Laroche, Mr. Joseph Philippe Lemercier        C       2
                                   Mallet, Mr. Albert        C       2
                                Mallet, Master. Andre        C       2
                          Duran y More, Miss. Asuncion        C       2
                 Abelson, Mrs. Samuel (Hannah Wizosky)        C       2
```

```
#ZAINA RAHEEN A P #241801328
# 13. Passengers who had more than 2 siblings/spouses
#     aboard (SibSp > 2)
# ------------------------------------------------------------

sibsp_more2 = df[df["SibSp"] > 2]
print("\n13. Passengers with SibSp > 2:")
print(sibsp_more2[
    ["Name", "SibSp"]
].to_string(index=False))
```

```
13. Passengers with SibSp > 2:
                                                Name  SibSp
                           Palsson, Master. Gosta Leonard      3
                                     Rice, Master. Eugene      4
                            Palsson, Miss. Torborg Danira      3
                           Fortune, Mr. Charles Alexander      3
                               Panula, Master. Juha Niilo      4
                         Goodwin, Master. William Frederick      5
                                  Skoog, Master. Harald      3
                            Andersson, Miss. Erna Alexandra      4
                                Goodwin, Miss. Lillian Amy      5
Backstrom, Mrs. Karl Alfred (Maria Mathilda Gustafsson)      3
                                Fortune, Miss. Mabel Helen      3
                          Andersson, Miss. Ellis Anna Maria      4
                                 Sage, Master. Thomas Henry      8
                               Panula, Master. Eino Viljami      4
                                     Rice, Master. Arthur      4
                           Lefebre, Master. Henry Forbes      3
                            Sage, Miss. Constance Gladys      8
                     Asplund, Master. Clarence Gustaf Hugo      4
                                   Sage, Mr. Frederick      8
                              Lefebre, Miss. Mathilde      3
                          Asplund, Miss. Lillian Gertrud      4
                         Asplund, Master. Edvin Rojj Felix      4
                                 Panula, Mr. Ernesti Arvid      4
                                       Rice, Master. Eric      4
                                 Sage, Mr. George John Jr      8
                            Fortune, Miss. Alice Elizabeth      3
                                Palsson, Miss. Stina Viola      3
                           Goodwin, Master. Sidney Leonard      5
                                       Lefebre, Miss. Ida      3
                           Goodwin, Master. Harold Victor      5
                                  Lefebre, Miss. Jeannie      3
                    Andersson, Miss. Ingeborg Constanzia      4
                       Andersson, Miss. Sigrid Elisabeth      4
                                   Skoog, Miss. Mabel      3
                            Skoog, Miss. Margit Elizabeth      3
                              Goodwin, Mr. Charles Edward      5
                                 Panula, Mr. Jaako Arnold      4
               Renouf, Mrs. Peter Henry (Lillian Jefferys)      3
                               Rice, Master. George Hugh      4
                                 Sage, Miss. Stella Anna      8
                         Andersson, Miss. Ebba Iris Alfrida      4
                              Skoog, Master. Karl Thorsten      3
                               Panula, Master. Urho Abraham      4
                                   Sage, Mr. Douglas Bullen      8
                    Andersson, Master. Sigvard Harald Elias      4
                          Sage, Miss. Dorothy Edith "Dolly"      8
```

```
#ZAINA RAHEEN A P #241801328
# 14. Passengers who did not survive and had no family
#     aboard (SibSp = 0 and Parch = 0)
```

```
# ------------------------------------------------------------

no_survive_no_family = df[
    (df["Survived"] == 0) &
    (df["SibSp"] == 0) &
    (df["Parch"] == 0)
]
print("\n14. Passengers Who Did Not Survive and Had No Family:")
print(no_survive_no_family[
    ["Name", "Survived", "SibSp", "Parch"]
].to_string(index=False))
```

```
                           Mack, Mrs. (Mary)         0      0      0
                            Elias, Mr. Dibo          0      0      0
       Myhrman, Mr. Pehr Fabian Oliver Malkolm       0      0      0
                           Tobin, Mr. Roger          0      0      0
                     Kilgannon, Mr. Thomas J         0      0      0
                      Long, Mr. Milton Clyde         0      0      0
                             Ali, Mr. William        0      0      0
            Harmer, Mr. Abraham (David Lishin)       0      0      0
                  Guggenheim, Mr. Benjamin           0      0      0
                  Keane, Mr. Andrew "Andy"           0      0      0
                       Gaskell, Mr. Alfred           0      0      0
                  Hoyt, Mr. William Fisher           0      0      0
                     Dantcheff, Mr. Ristiu           0      0      0
                        Otter, Mr. Richard           0      0      0
                Ibrahim Shawah, Mr. Youseff          0      0      0
                       Ponesell, Mr. Martin          0      0      0
                  Johansson, Mr. Karl Johan          0      0      0
                     Andrews, Mr. Thomas Jr          0      0      0
             Pettersson, Miss. Ellen Natalia         0      0      0
                         Meyer, Mr. August           0      0      0
                  Alexander, Mr. William             0      0      0
                       Lester, Mr. James             0      0      0
                  Slemen, Mr. Richard James          0      0      0
                 Tomlin, Mr. Ernest Portage          0      0      0
                          Fry, Mr. Richard           0      0      0
                Heininen, Miss. Wendla Maria         0      0      0
            Holm, Mr. John Fredrik Alexander         0      0      0
             Reuchlin, Jonkheer. John George         0      0      0
                           Flynn, Mr. John           0      0      0
                              Lam, Mr. Len           0      0      0
                           Saad, Mr. Amin            0      0      0
                  Augustsson, Mr. Albert             0      0      0
                  Allum, Mr. Owen George             0      0      0
                        Pasic, Mr. Jakob             0      0      0
                     Sirota, Mr. Maurice             0      0      0
               Alhomaki, Mr. Ilmari Rudolf           0      0      0
                  Mudd, Mr. Thomas Charles           0      0      0
                Lemberopolous, Mr. Peter L           0      0      0
                       Culumovic, Mr. Jeso           0      0      0
                       Abbing, Mr. Anthony           0      0      0
                        Markoff, Mr. Marin           0      0      0
                       Svensson, Mr. Johan           0      0      0
                          Razi, Mr. Raihed           0      0      0
                  Gill, Mr. John William             0      0      0
        Roebling, Mr. Washington Augustus II         0      0      0
                van Melkebeke, Mr. Philemon          0      0      0
                          Balkic, Mr. Cerin          0      0      0
                  Carlsson, Mr. Frans Olof           0      0      0
                Vander Cruyssen, Mr. Victor          0      0      0
              Gustafsson, Mr. Alfred Ossian          0      0      0
                       Petroff, Mr. Nedelio          0      0      0
                         Laleff, Mr. Kristo          0      0      0
                         Markun, Mr. Johann          0      0      0
               Dahlberg, Miss. Gerda Ulrika          0      0      0
              Banfield, Mr. Frederick James          0      0      0
                     Sutehall, Mr. Henry Jr          0      0      0
                      Montvila, Rev. Juozas          0      0      0
                        Dooley, Mr. Patrick          0      0      0
```

```
#ZAINA RAHEEN A P #241801328
# 15. Find the top 5 oldest passengers who survived
# ------------------------------------------------------------

oldest_survived = df[
    df["Survived"] == 1
].sort_values(
    by="Age",
    ascending=False
).head(5)
print("\n15. Top 5 Oldest Passengers Who Survived:")
print(oldest_survived[
    ["Name", "Age", "Survived"]
].to_string(index=False))
```

```
15. Top 5 Oldest Passengers Who Survived:
```

```
                                     Name  Age  Survived
     Barkworth, Mr. Algernon Henry Wilson 80.0         1
        Andrews, Miss. Kornelia Theodosia 63.0         1
                   Turkula, Mrs. (Hedwig) 63.0         1
                       Harris, Mr. George 62.0         1
Stone, Mrs. George Nelson (Martha Evelyn) 62.0         1
```

```
#ZAINA RAHEEN A P #241801328
# 16. List passengers who paid zero fare
# ------------------------------------------------------------

zero_fare = df[df["Fare"] == 0]

print("\n16. Passengers Who Paid Zero Fare:")
print(zero_fare[
    ["Name", "Fare"]
].to_string(index=False))
```

```
16. Passengers Who Paid Zero Fare:
                          Name  Fare
             Leonard, Mr. Lionel   0.0
           Harrison, Mr. William   0.0
    Tornquist, Mr. William Henry   0.0
     Parkes, Mr. Francis "Frank"   0.0
 Johnson, Mr. William Cahoone Jr   0.0
  Cunningham, Mr. Alfred Fleming   0.0
           Campbell, Mr. William   0.0
Frost, Mr. Anthony Wood "Archie"   0.0
             Johnson, Mr. Alfred   0.0
   Parr, Mr. William Henry Marsh   0.0
      Watson, Mr. Ennis Hastings   0.0
            Knight, Mr. Robert J   0.0
          Andrews, Mr. Thomas Jr   0.0
                Fry, Mr. Richard   0.0
 Reuchlin, Jonkheer. John George   0.0
```

```
#ZAINA RAHEEN A P #241801328
# 17. Split the dataset for training and testing
# ------------------------------------------------------------

# Features and target
X = df.drop("Survived", axis=1)
y = df["Survived"]

# Select numerical columns for a simple train-test split
X_numeric = df[
    ["Pclass", "Age", "SibSp", "Parch", "Fare"]
]

# Split into 80% training and 20% testing
X_train, X_test, y_train, y_test = train_test_split(
    X_numeric,
    y,
    test_size=0.2,
    random_state=42
)

print("\n17. Train-Test Split:")
print("Training data:", X_train.shape)
print("Testing data :", X_test.shape)
print("Training target:", y_train.shape)
print("Testing target :", y_test.shape)
```

```
17. Train-Test Split:
Training data: (712, 5)
Testing data : (179, 5)
Training target: (712,)
Testing target : (179,)
```

```
#ZAINA RAHEEN A P #241801328
# FINAL DATASET
# ------------------------------------------------------------

print("\nFinal Dataset:")
print(df.head())

print("\nMissing Values After Preprocessing:")
print(df.isnull().sum())
```

```
Final Dataset:
```

```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare    Cabin Embarked
0      0         A/5 21171   7.2500  Unknown        S
1      0          PC 17599  71.2833      C85        C
2      0  STON/O2. 3101282   7.9250  Unknown        S
3      0            113803  53.1000     C123        S
4      0            373450   8.0500  Unknown        S

Missing Values After Preprocessing:
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          0
Embarked       0
dtype: int64
```